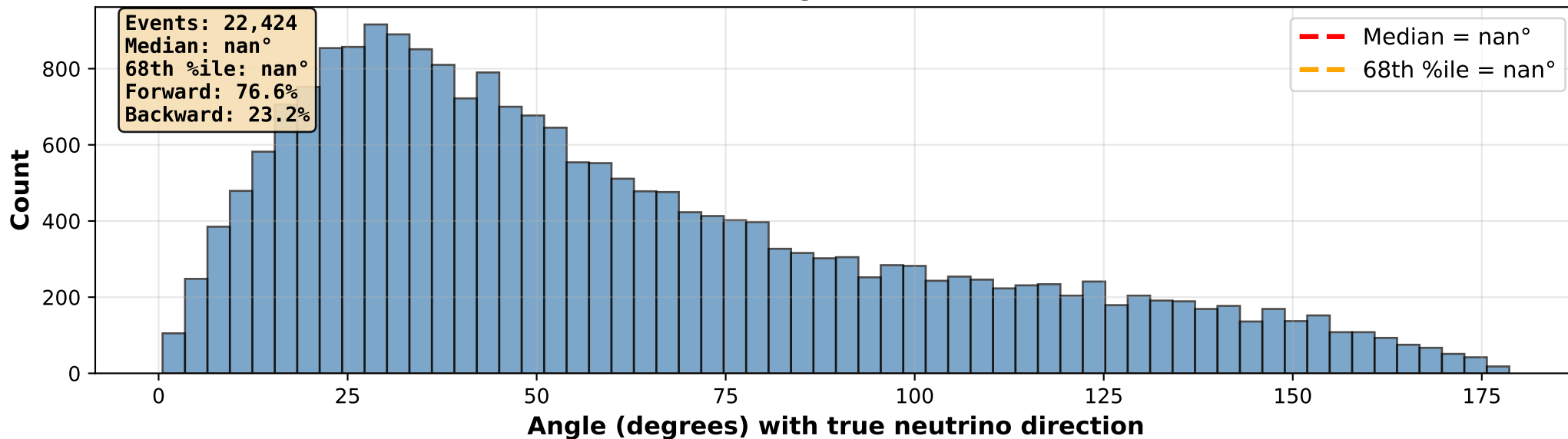


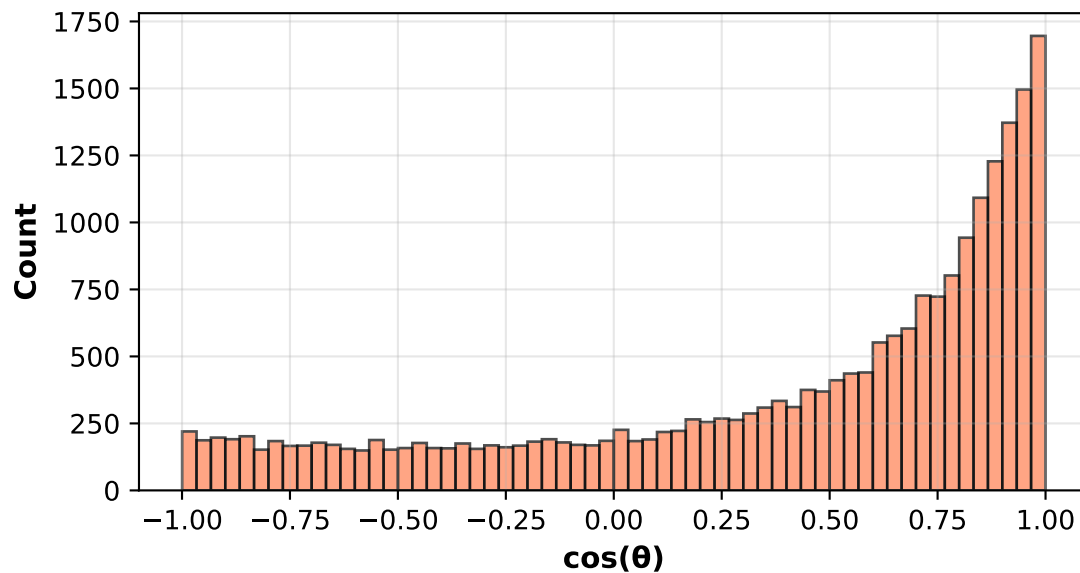
# ED Model + MCMC Pointing Performance

*Overall Results (561 categories, 22,424 events)*

## Overall Angle Distribution



## Cosine Distribution



### SUMMARY STATISTICS

Total Events: 22,424  
Successful Categories: 561

#### Angle Distribution:

Median: nan°  
Mean: nan°  
68th percentile: nan°  
Std deviation: nan°

#### Pointing Direction:

Forward (cos > 0): 17,176 (76.6%)  
Backward (cos < 0): 5,208 (23.2%)

### COMPARISON TO BASELINE

Best possible (true e<sup>-</sup> dirs):  
Angle 68% = 4.4°

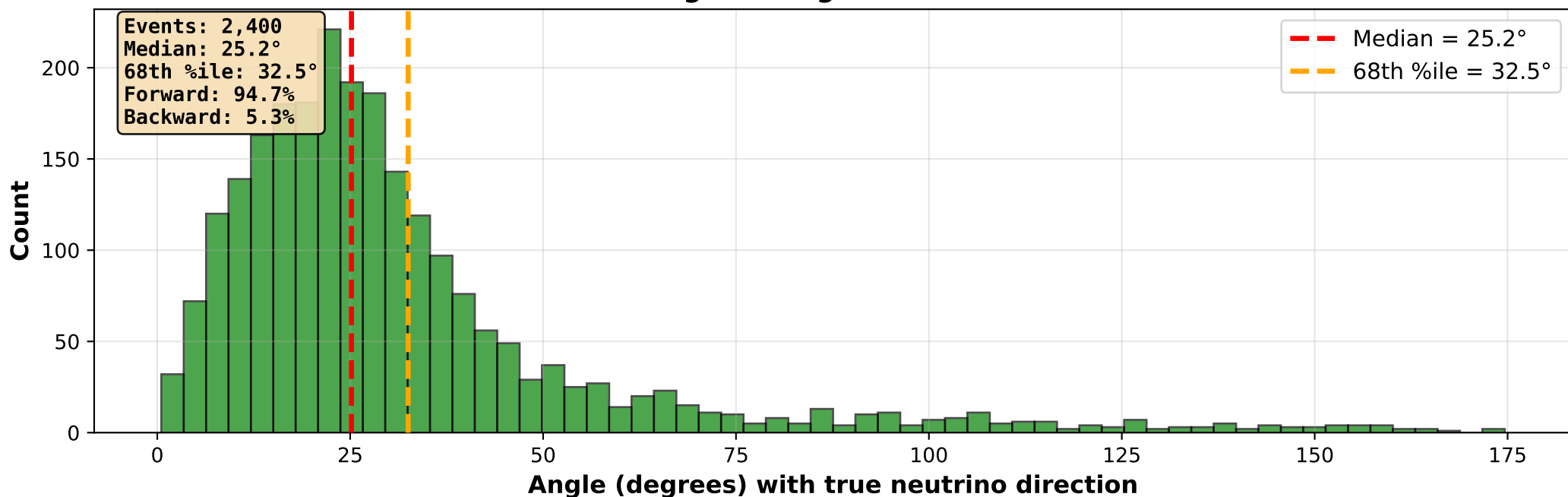
Current (ED model):  
Angle 68% = nan°

# ED Model + MCMC Pointing Performance

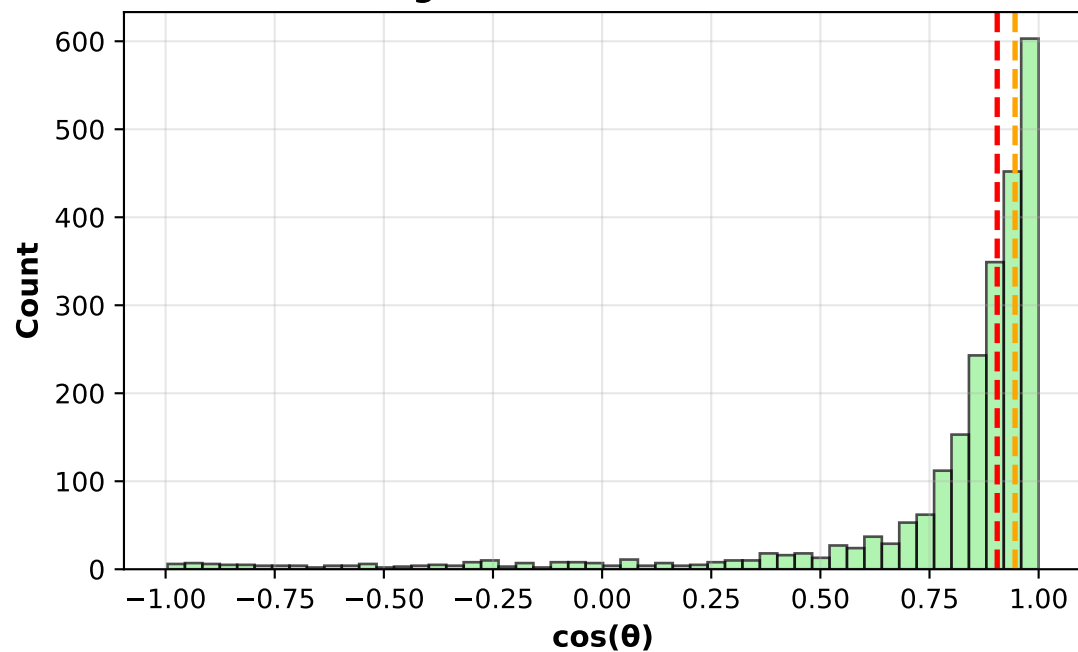
*Z-aligned (within  $\pm 30^\circ$  of axis)*

60 categories, 2,400 events

## Z-aligned: Angle Distribution



## Z-aligned: Cosine Distribution

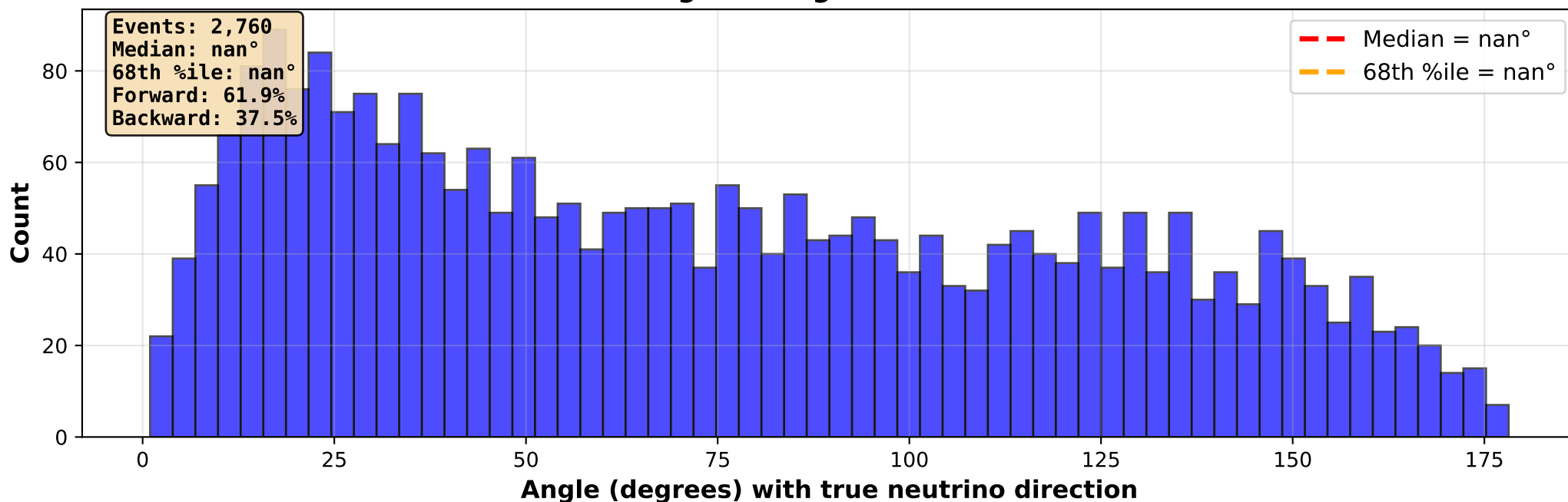


# ED Model + MCMC Pointing Performance

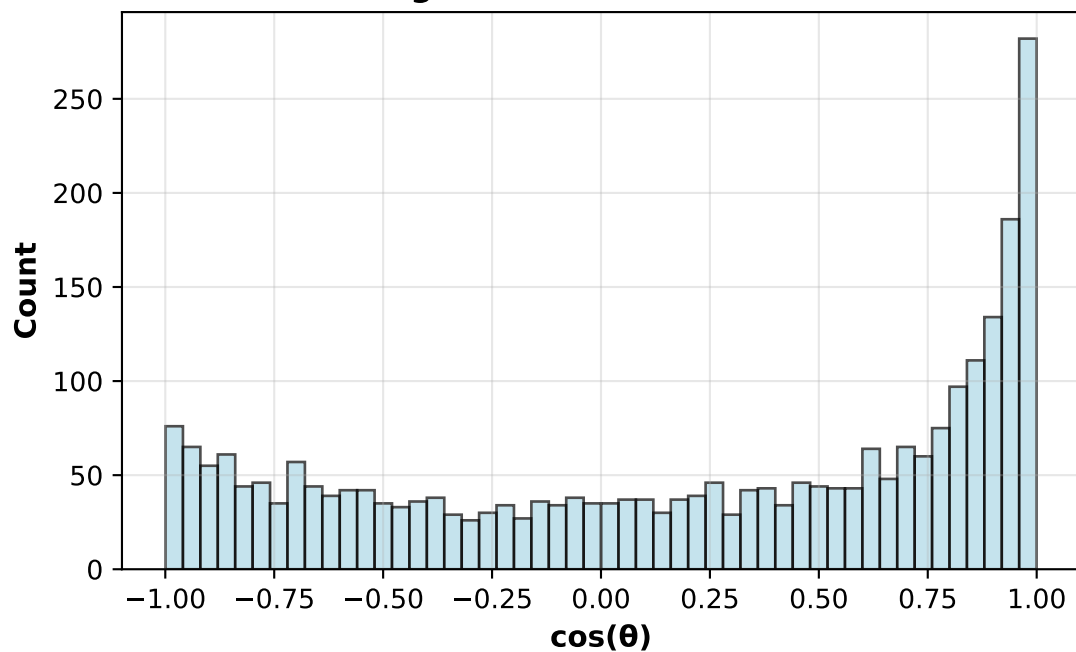
*X-aligned (within  $\pm 30^\circ$  of axis)*

69 categories, 2,760 events

## X-aligned: Angle Distribution



## X-aligned: Cosine Distribution

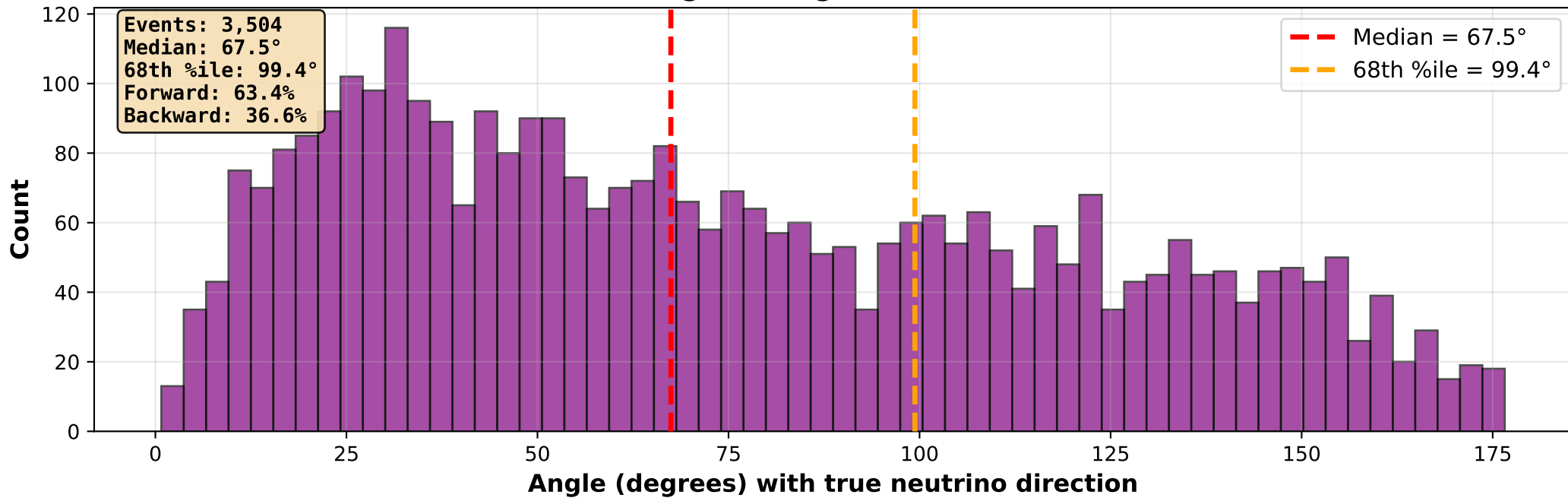


# ED Model + MCMC Pointing Performance

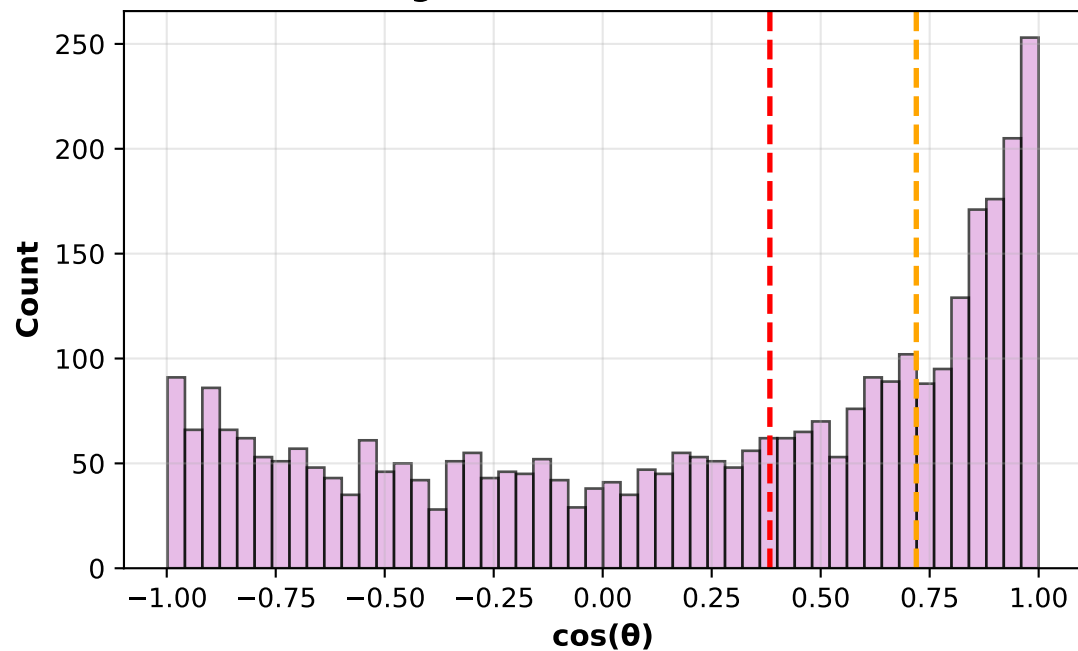
*Y-aligned (within  $\pm 30^\circ$  of axis)*

88 categories, 3,504 events

**Y-aligned: Angle Distribution**



**Y-aligned: Cosine Distribution**

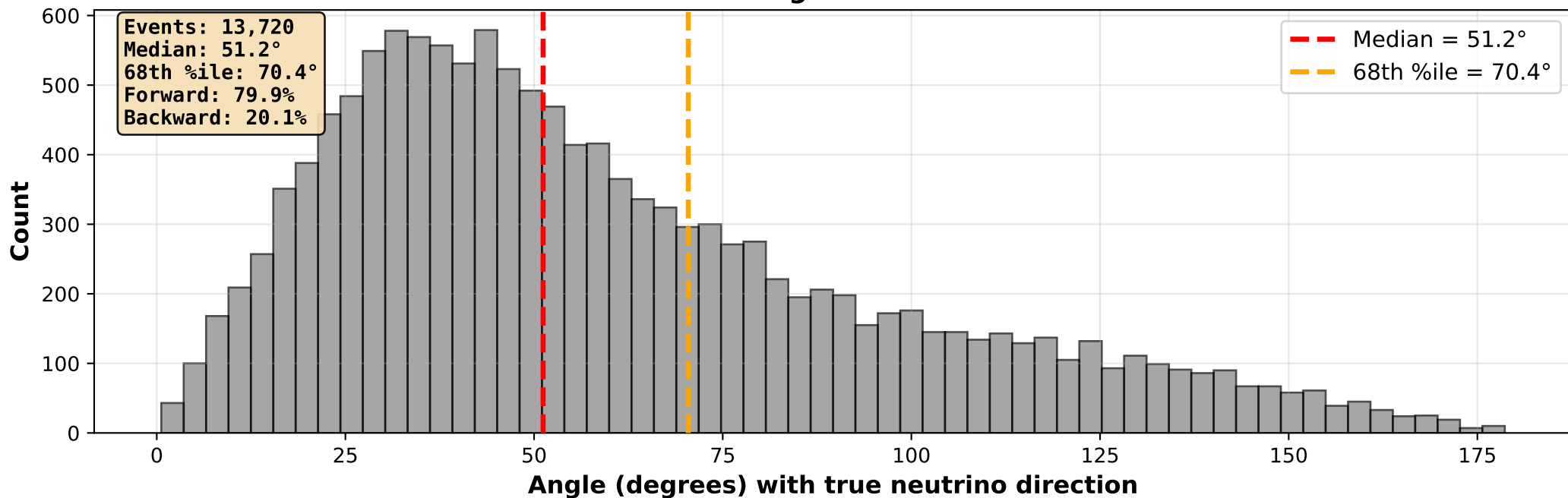


# ED Model + MCMC Pointing Performance

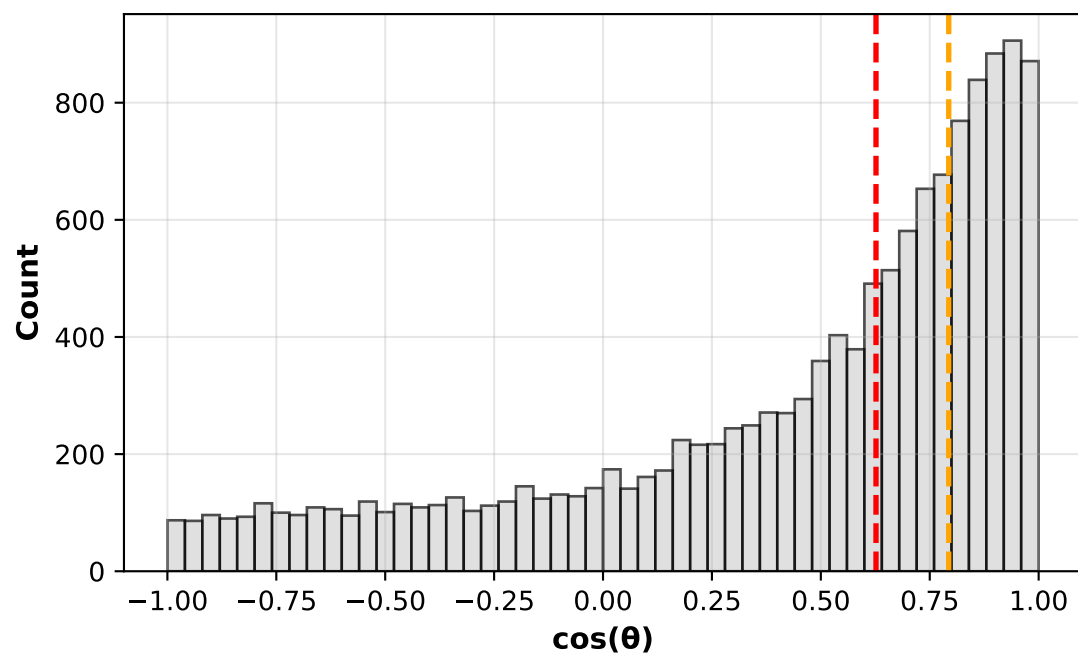
*Other (not aligned with any axis)*

343 categories, 13,720 events

**Other: Angle Distribution**

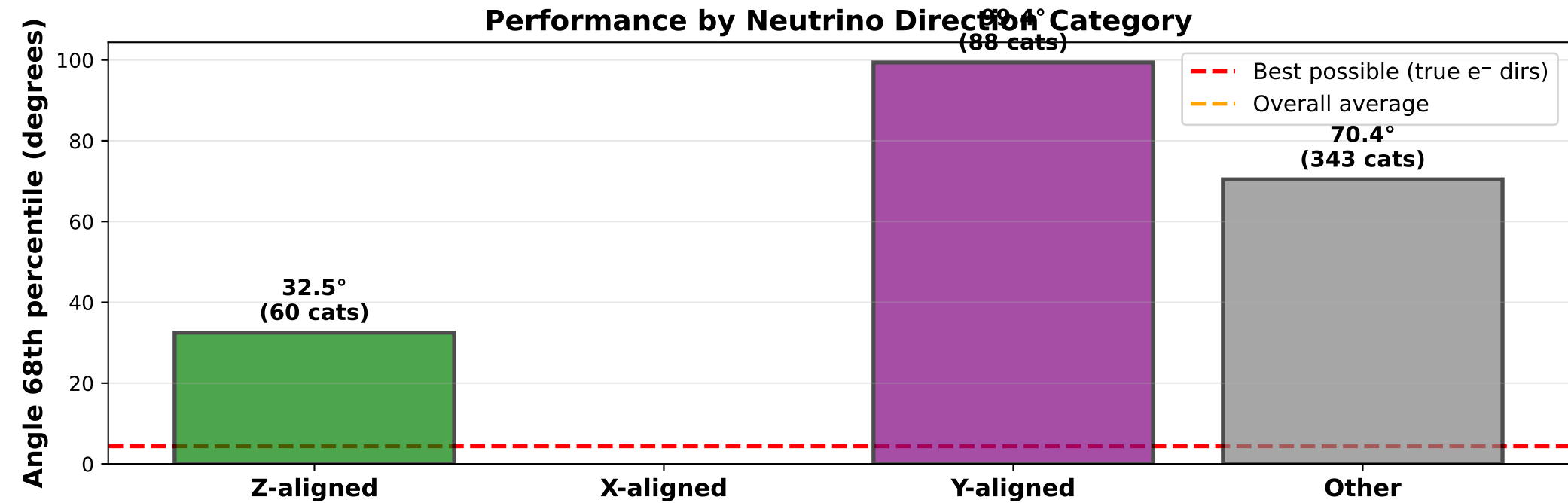


**Other: Cosine Distribution**



# ED Model + MCMC Pointing Performance

*Performance Comparison by Neutrino Direction*



| Performance Summary by Direction |        |          |           |         |          |
|----------------------------------|--------|----------|-----------|---------|----------|
| Category                         | N_cats | N_events | Angle 68° | Median° | Forward% |
| Z-aligned                        | 60     | 2,400    | 32.5°     | 25.2°   | 94.7%    |
| X-aligned                        | 69     | 2,760    | nan°      | nan°    | 61.9%    |
| Y-aligned                        | 88     | 3,504    | 99.4°     | 67.5°   | 63.4%    |
| Other                            | 343    | 13,720   | 70.4°     | 51.2°   | 79.9%    |
| Overall                          | 561    | 22,424   | nan°      | nan°    | 76.6%    |

| Key Findings                                           |  |
|--------------------------------------------------------|--|
| • Performance strongly depends on neutrino direction:  |  |
| - Z-aligned: 32.5°                                     |  |
| - X-aligned: nan°                                      |  |
| - Y-aligned: 99.4°                                     |  |
| - Other: 70.4°                                         |  |
| • Best possible (true e <sup>-</sup> directions): 4.4° |  |
| • Current gap: Δangle = nan°                           |  |